HS Data Science 26

Session 04: Scientific Working III — Sciometrics, Research Culture & Organisation

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Session Overview

Sciometrics & Research Quality

- measuring science: h-index, impact factor
- ranking venues: CORE, Scimago, DBLP
- predatory publishers and how to spot them
- open access and reproducibility

Research Culture

- the replication crisis and what it means for ML
- open science principles
- research ethics and AI-specific concerns

Organising Yourself

- Getting Things Done (GTD) for researchers
- weekly review as a habit
- tools: Zotero, Obsidian, Git research diary
- time-blocking and deep work

> Good science requires two things in equal measure: rigorous methods **and** sustainable work habits. Neither alone is enough.

Measuring Science — Sciometrics

> **Guiding Question:** How is scientific impact measured — and why does it matter which venue you publish in?

The h-index

Source: Hirsch 2005; Garfield 1955

Definition

A researcher has h-index **h** if exactly **h** of their papers have been cited at least **h** times (Hirsch 2005).

Example

Paper	Citations
P1	42
P2	17
P3	9
P4	6
P5	2

Strengths

- combines productivity and impact in one number
- robust to one-off highly-cited outliers
- widely used by hiring committees and funding agencies

Limitations

- favours quantity over quality of individual papers
- heavily field-dependent (ML h-indices are high; humanities are low)
- accumulates over time — disadvantages early-career researchers
- does not account for author order or self-citations

> Never compare h-indices across disciplines. An h-index of 15 means very different things

h-index = **4** (P1–P4 each have ≥ 4 cites)

Where to check

Google Scholar profile · Semantic Scholar · Publish or Perish (free)

> in physics vs. computer science vs. literary studies.

Impact Factor & Journal Metrics

Impact Factor (IF)

Average citations per paper published in a journal over the last 2 years.

$$IF = \frac{\text{citations to papers from years } t{-}1,, t{-}2}{\text{papers published in years } t{-}1,, t{-}2}$$

Published annually by Clarivate (Journal Citation Reports).

CiteScore (Scopus)

Similar idea, but uses a 4-year window — somewhat more stable.

SJR (Scimago)

Weighted by prestige of the citing journal — analogous to PageRank for journals.

What it tells you

- a rough signal of journal prestige within a field
- useful when selecting where to submit

What it does NOT tell you

- quality of any individual paper
- relevance to your specific research question
- reproducibility or rigour of published work

For Computer Science

Conference proceedings often matter more than journals. Many top venues (NeurIPS, ICML, ACL, WWW) have no impact factor — rank instead by CORE ranking or community acceptance.

> Impact Factor is a journal-level metric. It says nothing about whether a specific paper is worth reading.

Ranking Research Venues — CORE

Source: Research & Australasia 2023; Beall 2012

CORE Conference Ranking

The Computing Research and Education Association (CORE) ranks CS conferences:

Rank	Meaning
A*	flagship, top tier worldwide
A	excellent, highly selective
B	good, respected
C	peer-reviewed, limited impact
Unranked	no quality signal

Check at: portal.core.edu.au/conf-ranks

How to use it

When reading a paper: check venue rank to calibrate how much trust to extend before diving in.

When submitting: target A*/A venues even if rejection is likely — reviewers' feedback is educational and forces quality.

Warning: Predatory Venues

Signs of a predatory conference or journal:

- unsolicited invitation email promising fast review
- vague scope ("multidisciplinary", "all sciences")
- no indexing in DBLP, ACM DL, IEEE Xplore, Scopus
- excessive article processing charges with no clear affiliation
- check at: **bealslist.net**

For journals: Scimago (SJR) Q1–Q4 quartiles per discipline

Publishing in a predatory venue harms your reputation and wastes time. Always verify before submitting.

Open Science & Reproducibility

The Replication Crisis

Many published results — particularly in psychology and medicine — fail to replicate when experiments are repeated. ML is not immune:

- code not released → results unverifiable
- hyperparameter tuning hidden → baselines cherry-picked
- dataset leakage → inflated metrics
- random seeds not reported → variance hidden

Open Science Principles (UNESCO 2021)

1. Open data — share datasets where ethically possible
2. Open code — publish reproducible implementations
3. Open methods — describe setup completely
4. Open peer review — transparency in the review process

Practical Steps for AI/ML Papers

- release code on GitHub with a clear README
- report all hyperparameters and hardware specs
- use fixed random seeds, report variance across multiple runs
- publish to arXiv pre-print before or alongside journal submission
- use **Papers With Code** to link results to implementations
- use model cards and datasheets for datasets (Mitchell et al., 2019)

Reproducibility Checklists

NeurIPS and ICML now require reproducibility statements. Read them before writing — they are a free checklist of what reviewers will check.

Open, reproducible science is increasingly a requirement, not a bonus. Build the habit early.

Research Culture & Ethics

Research Ethics in AI

Fundamental Principles

- **Honesty:** report results as they are, including negative results
- **Transparency:** describe methods fully enough for others to replicate
- **Accountability:** acknowledge limitations and failure modes
- **Non-maleficence:** consider harms of the research and its applications

Specific AI Concerns

- bias in training data propagates to model outputs
- benchmark saturation creates false progress signals
- dual-use: the same model can help and harm
- environmental cost: large model training emits significant CO_2

Academic Integrity

Plagiarism: using others' text or ideas without attribution — zero tolerance, automated detection (iThenticate, Turnitin).

Fabrication / falsification: inventing or manipulating data — career-ending, potentially criminal.

Authorship: every named author must have made a substantial intellectual contribution. Gift authorship and ghost authorship are both violations.

AI-assisted writing: LLMs may be used as a writing aid, but you are responsible for every claim. Disclose use per venue policy.

When in doubt: cite, disclose, and describe. The cost of over-attribution is zero; the cost of under-attribution can end a career.

Organising Yourself — GTD for Researchers

> **Guiding Question:** How do you manage a research project that has no clear "done" and spans months?

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Getting Things Done (GTD) — Core Workflow

Source: Allen 2001

The Five Steps

1. **Capture** — collect every commitment, idea, task, and worry into a trusted inbox (notebook, phone, email folder)
2. **Clarify** — process each item: is it actionable? what is the next physical action?
3. **Organise** — put it in the right bucket: next-actions list, project list, waiting-for, someday/maybe, calendar
4. **Reflect** — weekly review: scan all lists, empty inbox, check project progress
5. **Engage** — choose what to work on based on context, energy, and priority

The Research-Specific Buckets

Bucket	Examples
Next Actions	"read Smith 2024 abstract", "run ablation on seed 42"
Projects	"seminar paper on RAG", "reproduce BEIR baseline"
Waiting For	"supervisor feedback on draft §3", "GPU cluster queue"
Someday/Maybe	"explore diffusion models for tabular data"
Reference	Zotero library, Obsidian notes, Git repo

Your brain is for thinking, not for storing. Capture everything externally and trust your system.

The Weekly Review

Why it matters: Research has no natural stopping point. Without a regular review, tasks accumulate invisibly and projects stall.

Weekly Review Checklist

- [ ] Process and empty inbox (email, notes, paper pile)
- [ ] Review last week's calendar — anything that needs follow-up?
- [ ] Check next-actions lists — mark done, identify stuck items
- [ ] Review all active projects — does each have a clear next action?
- [ ] Review waiting-for list — send follow-up nudges if overdue
- [ ] Review someday/maybe — promote anything now relevant

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Time and place

- 30–45 minutes, same time every week
- Friday afternoon is ideal (fresh week starts clean on Monday)
- physical separation helps: close all tabs first

Tools that support this

Purpose	Tool
Task management	Obsidian Tasks, Notion, plain text
Literature	Zotero (with Obsidian plugin)
Code & experiments	Git (commit = natural checkpoint)
Notes	Obsidian, Logseq, Roam

- [ ] Check upcoming calendar — prepare for commitments
- [ ] Identify the one most important thing for next week

Purpose	Tool
Calendar	Google Calendar (block deep work)

> The weekly review is a habit, not a method. It takes four weeks to feel natural. Do not skip it.

Deep Work and Research Time

Source: Newport 2016

Cal Newport's Deep Work Framework

- **Deep work:** cognitively demanding tasks that push the limits of ability — writing, proving, designing experiments
- **Shallow work:** logistical tasks that can be done while distracted — email, administrative forms, formatting references

Research value comes almost exclusively from deep work.

Key principles

1. Schedule deep work blocks (2–4 hours minimum) in the calendar
2. Eliminate distractions during deep work: phone off, notifications off

Research Rhythms

Block	Example
Morning deep work	writing, analysis, proofs
Mid-day shallow	email, Slack, administrative
Afternoon deep work	coding, experiments
Evening	reading papers (lighter cognitive load)

Protecting Deep Work

- batch email to two fixed times per day
- say no to low-value meetings

3. Measure deep work hours per week as a leading indicator
4. Rest deeply between blocks — shallow work in between is fine

- use "office hours" concept for collaboration requests
- log deep work hours (even a tally mark) — what gets measured improves

> Most researchers get 1–2 hours of genuine deep work per day. Aim for 4–6. That gap is where careers diverge.

Research Tools Stack

Literature Management — Zotero

- free, open-source reference manager
- browser plugin captures papers in one click
- syncs PDF highlights and annotations
- exports BibTeX directly for LaTeX
- Zotero Connector + Better BibTeX = seamless workflow

Note-Taking — Obsidian

- plain Markdown files, no lock-in
- bidirectional links: paper notes link to concept notes link to project notes
- Zotero plugin: pulls metadata and annotation highlights automatically
- graph view reveals connection patterns in your reading

The literature note template

Code & Experiments — Git as Research Diary

Every commit message is a lab notebook entry:

```
feat: add BM25 baseline on MSMARCO dev split

Scores: MRR@10 = 0.187, Recall@100 = 0.857
Same as reported in Thakur et al. 2021 Table 3.
Confirms our reimplementation is correct.
Next: add dense retrieval (DPR).
```

ML Project Structure

```
project/
├──── data/            # raw (gitignored), processed
├──── notebooks/       # exploration only, not production
├──── src/             # importable modules
├──── experiments/     # one folder per experiment run
│     └──── 2026-05-12-bm25-baseline/
│           ├──── config.yaml
│           ├──── results.json
│           └──── README.md
├──── tests/
└──── README.md
```

> Your future self is a stranger. Write notes and

```
## Citation
Author (year). Title. Venue.

## Core Claim
One sentence.

## Evidence
How did they test it?

## Limitations & Gap
What does this leave open?

## Connection to My Work
```

commit messages for them.

Summary

Sciometrics

- h-index = personal impact indicator (field-dependent!)
- Impact Factor = journal-level metric (not paper-level)
- CORE ranking = venue quality for CS conferences
- Predatory venues: always verify before submitting
- Open science: share code, data, methods

Research Ethics

- honesty, transparency, accountability
- reproducibility is a professional obligation
- AI-specific: bias, dual-use, environmental cost
- plagiarism, fabrication, authorship norms

Organisation

- GTD: Capture → Clarify → Organise → Reflect → Engage
- Weekly review: 30 min every Friday, non-negotiable
- Deep work: 4–6 hours/day is the goal
- Tools: Zotero + Obsidian + Git = complete research stack

> **The researcher's compact:** Measure your impact without being ruled by it. Organise your work so that deep thinking gets protected time. Open your results so others can build on them.

Image Credits & References

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